

Nels Buhrley

Rexburg, Idaho · 949-456-1531 · nelsbuhrley@gmail.com
[linkedin.com/in/nelsbuhrley](https://www.linkedin.com/in/nelsbuhrley) · nelsbuhrley.github.io

EDUCATION

Brigham Young University - Idaho
Bachelor of Science in Physics
GPA: 3.93

Rexburg, ID
Expected July 2027

SKILLS

- **Languages & Tools:** C++17, Python, GNU Make, Git, Linux/SSH, SLURM, LaTeX, Mathematica.
- **Engineering & Lab:** Positron Annihilation Spectroscopy (PAS), detector calibration/troubleshooting, SOLIDWORKS (CSWA), custom electronics/sensor prototyping.
- **Scientific Python:** NumPy, SciPy, Pandas, Matplotlib, Plotly, argparse; built tested CLI and visualization pipelines for simulation data.
- **Numerical Methods:** RK4, Euler-Cromer, Velocity Verlet, finite differences, red-black SOR, Metropolis MCMC, Monte Carlo random walks, FFT-based spectral analysis.
- **Parallel & HPC:** OpenMP multithreading and SLURM job scheduling; executed parameter-sweep, grid based, and stochastic simulations for both embarrassingly parallel and tightly coupled applications.

SELECTED TECHNICAL PROJECTS

- **Computational Physics Portfolio (C++17, Python, OpenMP, SLURM):** Built 12 simulation projects spanning ODE/PDE solvers, statistical mechanics, and molecular dynamics; deployed large parameter sweeps on HPC resources and produced reproducible analysis/visualization outputs.
 - **Molecular Dynamics:** Built a 2D Lennard-Jones simulator with velocity Verlet integration. Implemented particle skins to optimize interactions from $O(N^2)$ to near $O(N)$ achieving a 250,000x performance gain.
 - **3D Ising Model** Implemented Metropolis Monte Carlo with checkerboard updates and precomputed energy tables; mapped magnetization across temperature-field space and executed SLURM sweeps on up to 128 CPUs.
 - **Overrelaxation in Electrostatics** Solved 3D Laplace equations with red-black Successive Over-Relaxation, tuned optimal relaxation factors, and benchmarked convergence/scaling on parallel hardware.
- **TOPAS 3D Visualization Tool (Python):** Developed a production-ready CLI that ingests TOPAS/Geant4 CSV outputs, supports Cartesian/spherical/cylindrical transforms, and exports [interactive HTML](#) plus publication-ready static plots.
- **evt2bin Event Converter (C++):** Built a high-throughput text-to-binary pipeline using varint timestamp encoding, zlib compression, and external merge sort for memory-bounded processing of large Coincidence Doppler Broadening event logs.

EXPERIENCE

BYU Department of Physics - Computational X-Ray Imaging
Research Assistant

Rexburg, ID
June 2026 – Present

- **Software Development:** Leveraged GPU-accelerated libraries and custom CUDA kernels to implement a high-performance, parallelized X-ray image reconstruction pipeline, achieving a 400x speedup over prior methods.

- **Documentation:** Documented the development process and results of the X-ray image reconstruction pipeline using a searchable wiki-style documentation to facilitate knowledge transfer and future development.

Research and Business Development Center

Rexburg, ID

Research Lead - Positron Annihilation Spectroscopy (PAS)

April 2024 – July 2024

- **Software Development:** Wrote software in C++ and Mathematica for the analysis and visualization of complex PAS data, automating previously manual interpretation tasks.
- **Materials Characterization:** Characterized atomic-level defects in crystalline materials using Doppler Broadening Spectroscopy to analyze material purity and structural integrity.
- **Experimental Optimization:** Established metrology standards for spectroscopy systems to guarantee high-resolution, repeatable data. This was critical for materials process control, enabling studies on molecular absorption of water vapor by microporous materials.
- **System Troubleshooting:** Collaborated to investigate inconsistencies in Lifetime Spectroscopy data; isolated issues to voltage variations in photomultiplier tubes and established calibration procedures to prevent recurrence.
- **Leadership:** Co-developed a streamlined onboarding program that reduced trainee ramp-up time from 50 to 16 hours, representing a 68% decrease in training overhead.

Wright Woods

Mission Viejo, CA

Fabrication Specialist

Dec 2023 – June 2025

- **Industrial Systems Integration:** Independently engineered and commissioned a \$5,000 Donaldson Torit 20-5 dust collection system; managed end-to-end technical specifications, logistics, and physical installation to optimize shop safety.
- **Process Engineering:** Analyzed shop workflows to architect two major manufacturing pipeline improvements, generating over \$10,000 in added production value through optimized resource allocation.
- **Client Consultation:** Consulted with 10+ clients individually, performing on-site design updates and modifications that increased customer satisfaction and project efficiency.

LEADERSHIP & SERVICE

Community Service Impact: 500+ cumulative volunteer hours across youth mentorship, food bank operations, cemetery service projects, and direct one-on-one community support.

The Church of Jesus Christ of Latter-day Saints

Saint Louis, MO

Full-Time Volunteer Representative

Nov 2021 – Nov 2023

- **Team Leadership & Community Coordination:** Mentored and coordinated a team of 6–8 volunteers while partnering with local leaders to direct service where community needs were greatest.
- **Direct Service Impact:** Provided recurring one-on-one support to individuals and families and contributed to food bank distribution plus civic projects, including cemetery cleanup and community upkeep.

Boy Scouts of America

Orange County, CA

Eagle Scout & Youth Volunteer

July 2016 – July 2020

- **Large-Scale Service Operations:** Partnered with staff to deliver meals three times daily for 150+ youth and adult participants while maintaining strict health and safety standards.
- **Eagle Project (Built Environment):** Planned and built concrete garden planters and donated them to a Title I elementary school to support student learning spaces and campus beautification.